

neutral rate of interest remains unchanged, actual inflation is equal to its (new) target (Verify from equation (17.14)). Thus, fixing a higher inflation target results only in a temporary reduction in the interest rate. It causes a short-run increase in aggregate demand and output. In the long run, rate of interest cannot diverge from its neutral level.

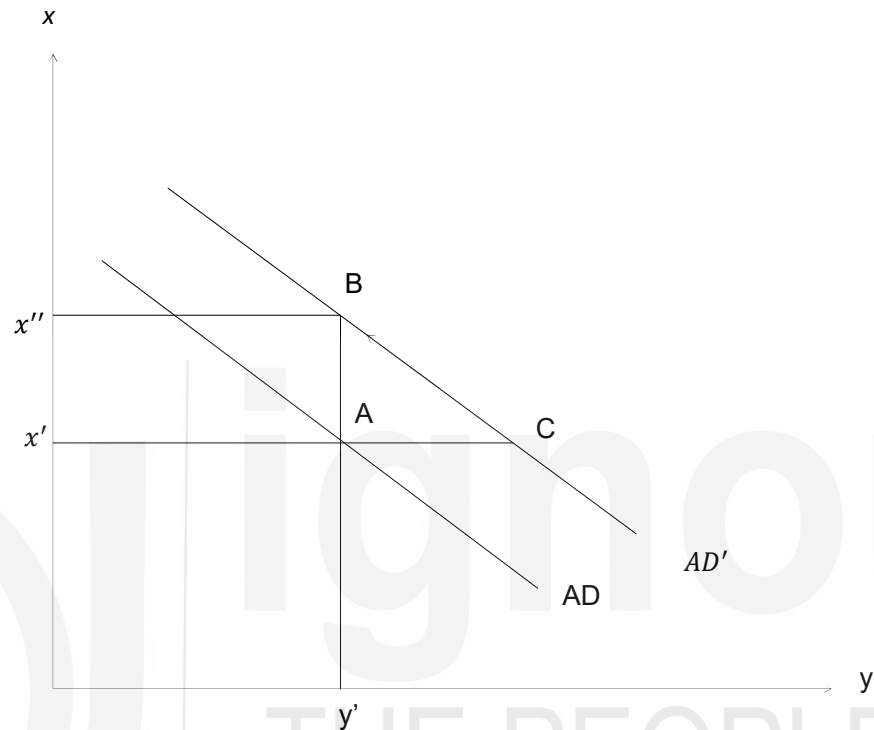


Fig. 17.3: Impact of a Higher Inflation Target

17.3.2 Impact of Expansionary Fiscal Policy

Let us assume that an increase in government expenditure causes the value of the parameter c to rise. This results in a higher aggregate demand for the given level of inflation and an outward shift of aggregate demand schedule from AD to AD' in Fig. 17.4. Thus, the economy moves from A to C . Inflation rate will now increase causing output to decrease along the AD' line. The economy would thus reach B . However, B in Fig. 17.4 is not an equilibrium point. This is because actual inflation rate now exceeds the target inflation rate. What should the Central Bank do to bring inflation back to its target? Notice that an increase in the value of parameter c increases the neutral rate of interest. As a result, r' must increase to ensure that inflation is at its target when output attains its target. Thus, the Central Bank increases the value of the parameter r' , which leads to a contraction of aggregate demand. The economy now moves from B to D in Fig. 17.4. From D , inflation falls and output increases along the AD line till we are back at the original equilibrium point, A .

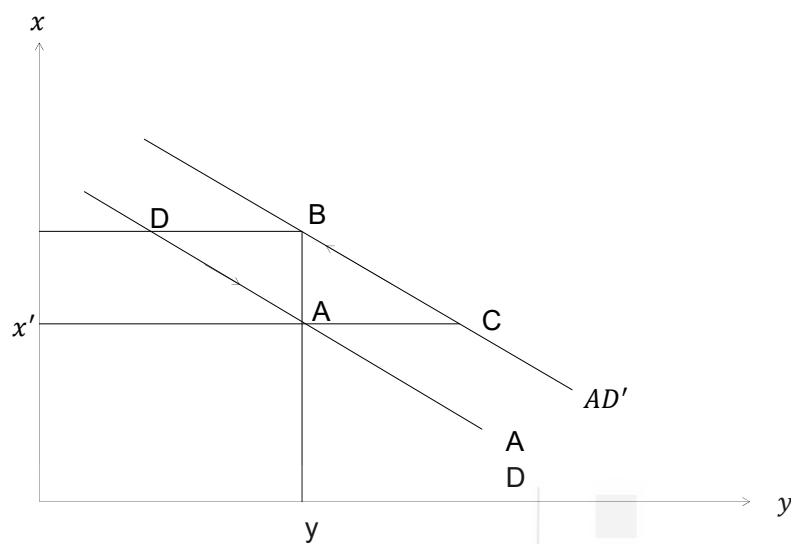


Fig. 17.4: Impact of Expansionary Fiscal Policy

An expansionary fiscal policy has no impact on either output or inflation in the economy (Notice that this view is in sharp contrast with the Keynesian view discussed in Unit 2). The only difference that an expansionary fiscal policy makes is to cause a higher interest rate at the same configuration of output and inflation (since the neutral rate of interest, which prevails at the equilibrium, has increased). Thus, expansionary fiscal policy results in a ‘crowding out’ of private expenditure. In the standard analysis of crowding out, higher public investment causes an excess of total (public and private) investment demand over savings supply. As a result, interest rate increases to equate investment with savings. In the new consensus view, interest rates are fixed by the Central Bank. However, an expansionary fiscal policy results in a higher inflation rate. The Central Bank is thus forced to push up the interest rate to maintain inflation at its target level.

We can sum up the main points of discussion so far. (i) Central Banks can play an important role in stabilising output at its natural level and inflation at some pre-defined target. (ii) Changes in monetary policy aimed at stimulating output above its natural level are bound to fail in the long run. If the Central Bank sets a higher inflation target, it only manages to achieve a higher rate of inflation with no improvement in output. (iii) The Central Bank may also try to reduce interest rate by fixing the rate of interest below the neutral rate of interest. But this policy will also fail to deliver any long run improvement in the output level (Why? What will be the impact of such a policy on inflation rate? Use the diagrams presented above). (iv) There is no space for an activist fiscal policy in the economy. The fiscal policy is also not to be entrusted with the role of controlling inflation and stabilising output levels. These objectives are to be pursued by a Central Bank

armed with an appropriate interest rate rule.

Check Your Progress 2

1. Describe the consensus view of monetary policy..

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2. Explain, how a higher inflation target influences the interest rate and output? How, these impacts are different from that of an expansionary fiscal policy?

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17.4 PATH DEPENDENCE IN MACROECONOMIC OUTCOMES

So far we have assumed that the level of natural output has been exogenously given. Many economists including M. Lavoie have pointed out that natural output level may actually depend on the level of aggregate demand and actual output. First, high actual output promotes learning-by-doing and contributes to improvement in labour productivity or reduction in per unit labour cost. Second, higher output levels induce more capital formation. Higher capital stock in turn improves labour productivity and reduces labour costs. Third, higher aggregate demand may encourage potential workers to enter the labour force and augment labour supply, which must also positively influence the level of natural output. All these factors must push up natural output level of the economy. Simply put, the capacity to produce depends on the amount actually produced. Therefore, we extend the model of the previous section by adding the following equation:

$$\Delta y' = f(y - y') \quad \dots (17.17)$$

Equation (17.17) indicates that the change in the level of natural output is directly or positively related to the output level and hence, the deviations of output. Whenever higher demand pushes actual output above its natural level, potential output increases. Let us assume that the Central Bank always fixes r' according to equation (17.15). Aggregate demand schedule given by equation (17.14) can then be re-written as:

$$y = y' + \left[\frac{da(x' - x)}{1 + bd} \right] \quad \dots (17.18)$$

On taking first difference, we obtain:

$$\Delta y = \Delta y' - \left[\frac{da\Delta x}{1 + bd} \right] + \left[\frac{da\Delta x'}{1 + bd} \right]$$

We can get the expressions for $\Delta y'$ from equation (17.17) and Δx from (17.13) to use them in the above equation. This gives:

$$\Delta y = \left[\frac{f - dae}{1 + bd} \right] (y - y') + \left[\frac{da\Delta x'}{1 + bd} \right] \quad \dots (17.19)$$

Inflation target x' is constant till the Central Bank changes its policy; $\Delta x'$ is therefore zero for a given policy stance of the Central Bank. Then, it is obvious from equations (17.17) and (17.19) that when $y = y'$ neither y nor y' will have any tendency to change. When the economy reaches the point where $y = y'$, it will tend to stay there. In Fig. 17.5, all points on the $y = y'$ line represent equilibrium combinations of actual output and natural output levels.

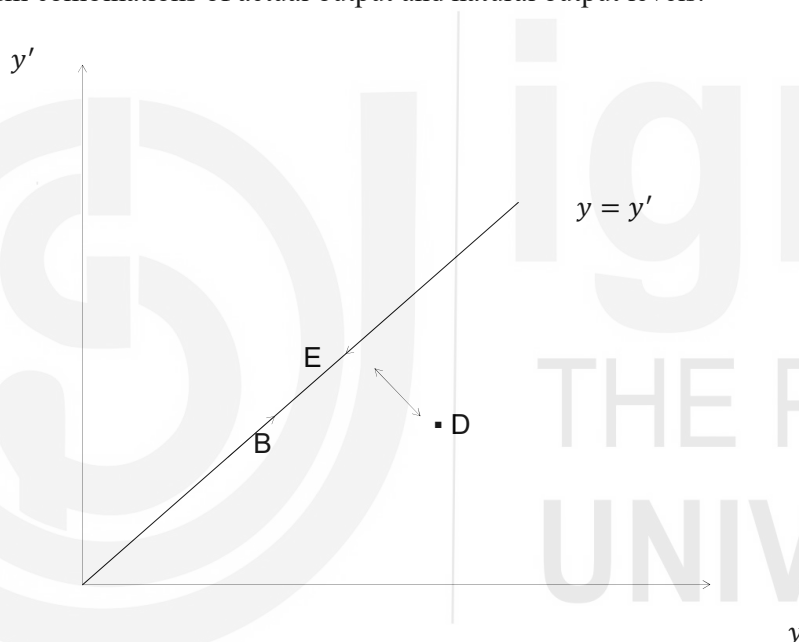


Fig. 17.5: Path Dependence

Suppose the economy is initially at B in Fig. 17.5. In the absence of any shock to the system, the economy will stay at B. Now, assume that the Central Bank changes its policy by declaring a higher inflation target. As a result $\Delta x'$ (and, consequently Δy) would become positive. The economy thus temporarily shifts from B to D in Fig. 17.5. The intuition is that a higher inflation target results in a reduction in interest rate that stimulates aggregate demand and output. As long as we assume that the square bracket term in equation (17.19) is negative, Δy would become negative once the economy reaches D. However, the economy will not settle at B again. This is because $y > y'$ at D so that y' will be increasing according to equation (17.17). The economy would thus move in the north-west direction till it reaches the new equilibrium, E. Both y and y' are higher at E compared to B. This model displays path dependence. In other word, there is no unique equilibrium point and the equilibrium we reach depends on the initial out-

of-equilibrium point we start from.

We have to be careful about one possibility in the above model. Factors that cause natural output to increase in response to high aggregate demand may take time to kick in. Such factors could be the entry of potential workers in the labour force, immigration of workers from other regions and improvements in labour productivity. If, in the meanwhile, excess demand pressures cause rapid acceleration of the inflation rate so that the Central Bank is also compelled to increase the rate of interest sharply, deviations from natural output level would be short-lived. We need to challenge this possibility because mainstream, following Friedman, believe that natural rate of unemployment has a knife-edge property. Any deviations from it would cause rapid acceleration or deceleration of inflation. In our model, high values of the parameter e (see equation (17.19)) may lead to such an outcome.

There are a number of studies that show that this is not the case. Most studies show that even if unemployment is reduced below its natural level for extended periods of time (natural unemployment corresponds to the natural output level; it is believed to exist because of frictions in the labour market) inflation rate increases only mildly. There is evidence on the shape of the short-run Phillips Curve, which shows some trade-off between unemployment and inflation (in the long run, as you know, there is no trade-off between unemployment and inflation and Phillips Curve is vertical at the natural rate of unemployment). However, the short-run Phillips Curve is usually concave. Thus, unemployment can be driven below its natural rate at a low cost in terms of inflation.

The idea of path dependence (also called hysteresis) in macroeconomic outcomes radically alters the conclusions of the new consensus macro model developed in the previous section. In the new consensus, as in the monetarist and new-classical views, actual output is pegged to an exogenously given natural output. Thus, fiscal and monetary policy interventions designed to increase the level of output are bound to fail. However, once we endogenise the potential or natural output of the economy, activist policies regain their potency. This is the main message of this section.

Check Your Progress 3

1. Explain the path dependence of outcomes in macroeconomics.

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17.5 LET US SUM UP

In this unit we discussed the rules vs. discretion debate. According to Milton

Friedman, Central Banks should follow a policy rule of constant growth in money supply. Other economists such as Robert Barro defined the rules as a commitment to zero/low inflation policy. For most of the new-Keynesians, the rules are shown in terms of equations that link a Central Bank's 'policy rate' to deviations from inflation and output targets. We observe that that much of the mainstream in economics prefers rules, however understood, to a discretionary use of policy by Central Banks.

Subsequently, we discussed the current consensus on monetary policy. In the consensus model, monetary policy plays a major role in fixing inflation rate at its pre-defined target. In other words, monetary policy can successfully provide the nominal anchor. Moreover, monetary policy, while successful in determining the rate of inflation, cannot influence the level of output, which is fixed at its natural level. In this sense, the new consensus is simply a rehash of the Friedmanite idea that market forces by themselves deliver full employment and monetary policy is neutral. At best, Central Banks can hope to reduce fluctuations of output around its natural level by following an appropriate interest rate policy.

Finally, we discussed the criticism of the new consensus model as it assumes that natural output level is fixed. Critics point out that natural output level may not be exogenously given. High aggregate demand (injected through monetary or fiscal means) may increase actual output as well as natural output levels and take the economy to a higher equilibrium.

17.6 ANSWR/HINTS TO CHECK YOUR PROGRESS EXERCISE

Check Your Progress 1

1. Refer to Section 8.2. Explain by using equations (17.1) to (17.10) and Fig. 17.1.
2. Refer to Fig. 17.1. Describe why the government should not change its policy stance afterwards.

Check Your Progress 2

1. The rate of interest is argued as the main instrument of the Central Bank to control inflation and output. You should explain the Taylor's rule, equations from (17.11) to (17.16) and Fig. 17.2. Refer to Section 8.3.
2. Refer to Section 8.3. Explain by using Fig. 17.3 and Fig. 17.4.

Check Your Progress 3

1. The natural output level may not be exogenously given. It may actually depend on the level of aggregate demand and actual output. First, high actual output promotes learning-by-doing and contributes to improvement in labour productivity or reduction in per unit labour cost. Also, higher output levels induce more capital formation. Higher capital stock in turn improves labour productivity and reduces labour costs. All these factors must push up natural output level of the economy. At the same time the high demand may encourage potential workers to enter the labour force and augment labour supply, which must also positively influence the level of natural output. Explain equations (17.17), (17.18) and (17.19), and Fig. 17.5.